Nepal College of Information Technology
Unit Test 2025-Fall

Program: Computer — Time: 2 Hours
Semester: V — FM: 70
Subject: Digital Signal Analysis and Processing — PM: 35

- ✓ *Candidates are requested to give their answer as far as practicable in their own words.*
- ✓ *The figure in the margin indicates the full marks*
- ✓ ***Attempt <u>ALL</u> question***

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.
Attempt all the questions.

1 a) A digital communication link carries binary code words representing samples of an input signal $x_a(t)=5\cos 600\Pi t+7\cos 800\Pi t$. The link is operated at 1000bits/sec and each input sample is quantized into 1024 different voltage levels. — 8

I. What is the sampling frequency and folding frequency?
II. What is the Nyquist rate for the signal $x_a(t)$?
III. What are the frequencies in the resulting discrete time signal x[n]?
IV. What is aliasing effect?

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b) Show that the impulse response of a BIBO stable LTI system is absolutely summable. — 7

2 a) .Determine Z-transform along with ROC of — 8

$$x(n)=[(1/2)^n + (3/4)^n]u(n-5)$$

b) An LTI system is described by the equation — 7

$$y(n) = x(n) + 0.81x(n-1) - 0.81x(n-2) - 0.45y(n-2)$$

Determine the transfer function of the system. Sketch the poles and zeros on the z-plane. Assess the stability.

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3 a) Define time-invariance and causality of a discrete time systems. 7

Determine whether the following discrete-time signals are periodic or not. If periodic, determine the fundamental period.

a. $\cos\left(\frac{n}{6}\right)\cos\left(\frac{n\pi}{6}\right)$

b. $\cos\left(\frac{\pi}{2}+0.3n\right)$

b) Find the circular convolution of two finite duration sequences using graphical method. 8

$$x_1(n)=\{1,2,-1,-2,3,1\}, x_2(n)=\{3,2,1\}$$

4 a) Find the response of a LTI system with input x(n)={9,**8**,7,6} to an impulse response h(n)={5,4,**8**,7},where bold letter denotes origin. Use graphical approach. 8

b) Determine the causal signal x(n) if its Z-transform X(z) is given by 7

$$X(z)=\frac{1+3z^{-1}}{1+3z^{-1}+2z^{-2}}$$

5 Write short notes on: (Any two) 5 x 2=10

a) Basics elements of DSP
b) Properties of Z transform
c) Causal and Non Causal systems

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National Academy of Science and Technology

(Affiliated to Pokhara University)

Dhangadhi, Kailali

First Term Examinations

Semester: VIII/V_Fall

Level: Bachelor
Program: B.E. Computer
Course: **Digital Signal Analysis &Processing**

Year : 2025
F.M. : 100
P.M. : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1 a) Given the analog signal: $x(t)= 2 \sin 300\pi t + 5 \sin 500\pi$
   i) Determine the minimum sampling rate required to avoid aliasing.
   ii) Suppose that the signal is sampled at the rate F_s =200 Hz. What is the discrete-time signal obtained after sampling?
   iii) What is the frequency $0< F<F_s/2$ of a sinusoid that yields samples identical to those obtained in part (II)? [8]

b) Explain the Basic elements of digital signal processing with the help of block diagram. [7]

2a) What are Discrete-time signals. Explain different elementary signals with its mathematical expression and signal representation. [7]
b) Perform the convolution sum of the following two sequences:
$x_1(n) = \{1, 2, 5, 6\}$ and $x_2(n) = \{2, 1, 0, 2\}$ [8]

3a) What is System. Explain different properties of system. [7]
b) Find the response of given discrete time signals
$x(n)= u(n)$ and $h(n)= a^n u[n]$

4a) Determine the z-transform of the signal
$x[n]=a^n u[n]$, $a>0$; where $u[n]$ is the unit-step function. [7]
b) Determine the z-transform and the region of convergence (ROC) for the signal: $x[n]= \cos(w_o n)\, u[n]$ [8]

5a) Determine the causal signal x[n] having Z-Transform [7]
$$X(z) = 1/\{(1-2z^{-1})(1-z^{-1})^2\}.$$

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b) Draw the correlation of the two discrete time signals
$x_1(n) = \{1, 2, 3, 4\}$ and $x_2(n) = \{0,2,4,6\}$ [8]

6a) Define z-transform. Explain Different properties of Z transform. [8]
b) Determine the z-transform and the region of convergence (ROC) for the signal: $x[n] = n^2 u[n]$ [7]

7. Write short notes on (any two) [2x5=10]
a. Region of Convergence
b. Energy signal and power signal
c. Digital Signal Processor vs Analog Signal Processor

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COSMOS COLLEGE
OF MANAGEMENT AND TECHNOLOGY

Term Test I

Date: 2082/09/30			
Level	BE	Full Marks	50
Programme	BCE	Time	
Semester	V	1.5 hrs	

Subject: - Digital Signal Analysis & Processing

- ✓ *Candidates are required to give their answers in their own words as far as practicable.*
- ✓ *Attempt All questions.*
- ✓ *The figures in the margin indicate **Full Marks**.*
- ✓ *Assume suitable data if necessary.*

1. a What are the advantages of digital signal processing over analog signal processing? Briefly explain the basic elements of digital signal processing with the help of block diagram. 8

   b The discrete time signal x[n] = 6.5cos(0.1πn) is quantized with a resolution, Δ = 0.01. How many bits are required in the A/D converter? If the maximum frequency that can be reconstructed from above signal is 500 Hz, determine the reconstructed continuous time signal. 7

2. a Obtain a linear convolution of the following two discrete time signals: $x[n] = \sum_{k=0}^{3} \partial[n-K]$ and $h[n] = 2^n[u[n] - u[n-3]]$. Also verify your answer. 8

   b Define FFT. Determine the FFT of the sequence $x(n) = \{1,2,3,4,5,6,7,8\} - 2\partial(n-4)$ 7

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3. a Define Z-transform and ROC. Find the inverse z transform of $X(Z) = \log(1 + az^{-1})$ 7

   b Determine which signals are periodic and compute fundamental period. 8

   i. $\cos\left(\frac{\pi n^2}{8}\right)$ ii. $\sin\left(\frac{3\pi}{2}\right)\cos\left(\frac{5\pi n}{4}\right)$

4. Write short notes: (any one) 5

   a Radix 2

   b FFT

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Nepal Engineering College

Assessment (Fall Semester)

Level: Bachelor
Program: Computer
Course: Digital Signal Analysis &Processing
Year/Part: III/I

Year: 2025
Full Marks: 100

- ✓ *Candidates are required to give their answers in their own words as far as practicable.*
- ✓ *Assume suitable data if necessary or missing.*
- ✓ *Attempt all the questions*

1 a Define elementary signals. Explain and draw different types of basic elementary signals. 7

b A digital communication link carries binary-coded words representing samples of an input signal 8
$x_a(t) = 4\cos 3500\pi t + \sin 4500\pi t$.
The link is operated at 42,000 bits/s and each input sample is quantized into 1024 different voltage levels.

a) What is the sampling frequency and folding frequency?
b) What is the Nyquist rate for this signal?
c) What is the discrete-time signal obtained after sampling?
d) What is the resolution Δ?

OR

Define discrete-time system. Determine whether or not the given signal $y[n] = nx[2-n^2]$ is
i) Linear ii) Causal iii) stable iv) time-invariant v) with memory or not?

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2 a State and prove time shifting and convolution properties of discrete-time Fourier transform. 7

b Consider an LTI system with impulse response $h[n] = \{1, \underset{\uparrow}{4}, 2, -1\}$ and input 8
$x[n] = \{1, 1, \underset{\uparrow}{4}, 13, 15\}$. Determine the output of the system, y[n] using graphical method.

3 a Define z-transform and ROC. Determine the z-transform and sketch the Roc of the following signal (use z-transform properties). 7
$x[n] = n^2 0.8^n u[n]$

b Draw the pole-zero plot and find the magnitude response of the following system: 8
$H(z) = \frac{1 + 2z^{-1}}{1 - 0.5z^{-1}}$

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4 a Consider the following system described by LCCD equation: 7
$y[n] = 0.6y[n-1] - 0.11y[n-2] + 0.3y[n-3] + x[n] + 2x[n-1] + 3x[n-3]$.
Draw the direct form- I and form-II structures.

b Determine the lattice coefficients corresponding to the FIR system with the system function 8

$$H(z) = 1 + \frac{7}{9}z^{-1} + \frac{3}{5}z^{-2} + \frac{1}{7}z^{-3}$$

5 a Design a lowpass Butterworth filter to meet the following specifications 8
Passband cutoff frequency: 200 Hz
Stopband cutoff frequency: 500 Hz
Passband ripple: 3 dB.
Stopband ripple: 21 dB.

OR

The z-transform of the signal is described as

$$X(z) = \frac{1 + 3z^{-1}}{(1 - 0.5z^{-1})(1 + 0.9z^{-1})}$$

What are the possible ROCs? Determine the inverse z-transform for causal signal.

b Differentiate FIR and IIR digital filters. 7

6 a Suppose we have two sequences x[n] and h[n] as follows: 7
$x[n] = \{1, 1, 4, -1, 7\}$; $h[n] = \{1, 4, 5, -2\}$.
Calculate circular convolution graphically.

b Find 8-point DFT of the sequence {12, 2, 5, 6, 0, -6, -5, -2} using radix-2 decimation in time domain. 8

7 Write short notes on (Any two): 5*2

a Causality and stability of LTI system

b Butterworth Filter

c Properties of z-transform

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Order of Filter N	Transfer function H(s) = 1/A(s) where A(s)
2	$(s^2+0.707s+1)$
3	$(s^2+s+1)(s+1)$
4	$(s^2+0.766s+1)(s^2+1.848s+1)$
5	$(s^2+0.618s+1)(s^2+1.618s+1)(s+1)$
6	$(s^2+0.518s+1)(s^2+4.414s+1)(s^2+1.932s+1)$
7	$(s^2+1.802s+1)(s^2+1.247s+1)(s^2+0.445s+1)(s+1)$

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United Technical College

Level: Bachelor | Semester: Fall | Year: 2025

Programme: BE | Full Marks: 50

Course: Digital Signal Analysis and Processing | Pass Marks: 25

Time: 1.5 Hours

Instructions:

Candidates are required to answer the questions in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt any three from 1-6 questions. Q.N.7 is compulsory.

Draw neat and labelled diagrams wherever necessary.

1. a. Define DSP. Explain the applications of DSP in the field of communication engineering. [7]

   b. A Digital communication link carries binary coded words representing samples of an input signal $x_a(t) = 3cos5000\pi t + 2sin6000\pi t$ Link is operated at 18,000 bits/s. and each input sample is quantized into 512 different voltage levels.

      i. What is the discrete time signal obtained after sampling?

      ii. What is the resolution? [8]

2. a. A discrete time Signal x(n) is applied to a discrete time LTI system with unit impulse h(n). Find output or response y(n) if given that

   $x(n) = a^n u(n)$, $0 < a < 1$ and $h(n) = u(n)$ [7]

   b. Define Z transform. Determine the Z transform of

   $x(n) = na^n u(n)$ [8]

3. a. Find the inverse z - transform of the system [7]

   $$X(z) = \frac{5 - 2z^{-1}}{2 - z^{-1} - 1.5z^{-1}}$$

   For the following ROC

   i. $|z| < 1$ ii. $|z| > 3$ iii. $1 < |z| < 3$

   b. Obtain the circular convolution of the following sequences [8]

   $x_1(n) = \{1, 2, 1, .1\}$ and $x_2(n) = \{4, 0, 2, 2\}$

5. a. Define machine learning. Explain the K-means clustering with an example. 7

   b. Explain backpropagation in neural networks. Why is the activation function important? 8

6. a. Explain the steps of fuzzification and defuzzification with examples. 7

   b. Define expert systems. Explain their architecture and components with a diagram. 8

7. Write short notes on(Attempt Any Two): 5

   a. Impact of AI on employment, privacy, and security

   b. Linear Regression

   c. CSP

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Pokhara Engineering College
Internal assessment

Level: Bachelor
Programme: BE Computer (V)
Course: Digital Signal Analysis and Processing

Year: 2025
Full Marks: 100
Time: 3 hrs.

Candidates are required to give their answer in their own words as far as practicable.
Figures in the margin indicate full marks.
Attempt all the questions.

1. a) What is equiripple filter? Write the Remes exchange algorithm for optimum equiripple filter design. 8

   b) Define signal processing. What are the advantages of DSP over ASP? 7

2. a) A digital communication link carries binary-coded words representing samples of an input signal $x_a(t) = 3cos600\pi t + 2cos1800\pi t$. The link is operated at 10,000 $bits/s$ and each input sample is quantized into 1024 different voltage levels.
   a. What are the sampling frequency and folding frequency?
   b. What is the Nyquist rate for the signal $x_a(t)$?
   c. What are the frequencies for the resulting discrete-time signal $x[n]$?

   b) Define Convolution. Obtain the necessary ad sufficient condition for stabie LTI system. 7

3. a) Determine the causal signal x(n) having the Z-transform
   $X(Z) = 1/[(1-2z^{-1})(1-z^{-1})]$ 8

   b) Define ROC. Explain the properties of ROC. 7

4. a) Design a lowpass filter to be used in an ADC-H(z)-DAC structure that will have a $-3\ dB$ cutoff of $30\pi\ rad/sec$ and an attenuation of 40 dB at $45\pi\ rad/sec$. The filter is required to have a linear phase. Use a sampling rate of 100 $samples/sec$. 8

   b) Design an FIR linear phase FIR filter using Kaiser window to meet the following specifications. 7

   $0.99 \le |H(e^{j\omega})| \le 1.01, \quad 0 \le |\omega| \le 0.4\pi$
   $0 \le |H(e^{j\omega})| \le 0.001, \quad 0.6\pi \le |\omega| \le \pi$

5. a) Obtain the cascade structure of the system with difference equation. 7
   $y[n] = 3/4\ y[n-1] - 1/8\ y[n-2] + x[n] - 1/3\ x[n-1]$

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$F_s = \frac{\text{bit rate}}{\text{bits sample}} = 1\text{KHz}$

$f_{min} = \frac{F_s}{2} = 500\text{Hz}$

$t = \Delta T = \frac{1}{F_s}$ sec

$F_{max} = 400\text{ Hz}$

$N.Q = 2F_{max} = 800\text{ Hz}$

b) Given a three stage lattice filter with coefficients $K_1 = \frac{1}{4}$, $K_2 = \frac{1}{4}$, $K_3 = 1/3$, determine the FIR filter coefficients for the direct-form structure. 8

6. a) Design a low pass discrete- time filter by applying impulse invariance to an approximate Butterworth continuous filter, if passband frequency is 0.5π radians and maximum deviation of 1dB below 0 dB gain in the passband. The maximum gain of -40 dB and frequency is 0.7π radians in the stopband. 8

b) Discuss the FIR filter design by IIM method. 7

7. Write short notes on the following **(any two):** 2×5

a) Linear convolution using Circular Convolution

b) Elementary discrete-time signals

c) Gibb's phenomenon

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National Academy of Science and Technology

Dhangadhi, Kailali

Pre-University Examination

Level: Bachelor | Semester – Fall | Year : 2025
Programme: BE | Full Marks : 100
Course: Digital Signal Analysis Processing | Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) A digital communication link carries binary-coded words representing samples of an input signal $X_a(t) = 5\cos 800\pi t + 3\cos 1200\pi t$. The link is operated at 20,000 bits/s and each input sample is quantized into 1024 different voltage levels. 8

   i) What is the sampling frequency and the folding frequency?

   ii) What is the Nyquist rate for the signal $X_a(t)$?

   iii) What are different frequencies in the resulting discrete-time signals x[n]?

   iv) What is the resolution Δ? 7

   b) Highlighting the key features of Digital Signal processing, briefly differentiate Digital Signal Processor with Analog Signal Processor.

2. a) Define LTI system. Mention and explain the properties of LTI system. 8

   OR 7

   Show that the multiplication of two DFT sequences results in circular convolution

   b) Obtain a linear convolution of the following two discrete-time signals:

   $x(n) = \{0,1,2,3\}$ and $h(n) = 2n[u(n)-u(n-3)]$

3. a) For the given system $y[n] = 0.7\, y[n-1] - 0.1\, y[n-2] + x[n]$, plot the pole zero diagram and magnitude response of the system 8

   b) Determine the causal signal x[n] having Z-Transform $X(z) = 1/\{(1-2z^{-1})(1-z^{-1})^2\}$. 7

4. a) Define DFT. Also compute 4-point DFT of the following sequence: 7

   $x(n) = \{1, 2, 3, 4\}$

   b) . Find 8- point DFT of the sequence. {0,0,0,0,1,1,1,1,} using radix 2 DITFFT. 8

5. a) Here is given a three stage lattice filter with coefficient $K_1 = 1/4$, 8

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$K_2 = ½$, and $K_3 = 1/3$, determine the FIR filter coefficient for the direct-form structure. 7

b) Determine the lattice coefficients corresponding to the FIR filter with system function and also draw the lattice structure.

$$H(Z) = 1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}.$$

6. a) Design an IIR filter by using Bi-linear transformation to meet the following specification $w_p = 0.25\pi$, $w_s = 0.55\pi$, $\alpha_s = 15dB$, $\alpha_p = 0.5dB$. 8

OR

Design a linear FIR filter using Kaiser window to meet the following

$$0.99 \leq |H(e^{j\omega})| \leq 1.01, for 0 \leq |\omega| \leq 0.19\pi$$

$$\leq |H(e^{j\omega})| \leq 0.01, for 0.21\pi \leq |\omega| \leq \pi$$

b) Draw the flowchart of Remez algorithm and explain it

OR

The transfer function of Analog filter is $Ha(s) = \frac{2}{(s+2)(s+4)}$ with T = 0.1 sec. Design the IIR digital filter by using impulse invariance m. 7/12

7. Write short notes on (Any Two): 5×2

a) Gibb's Phenomena

b) Cascade structure for FIR system

c) ROC of Z transform

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National Academy of Science and Technology

Dhangadhi, Kailali

Pre-University Examination

Level: Bachelor | Semester – Fall | Year : 2025
Programme: BE | Full Marks : 100
Course: Digital Signal Analysis Processing | Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) A digital communication link carries binary-coded words representing samples of an input signal $X_a(t) = 5\cos 800\pi t + 3\cos 1200\pi t$. The link is operated at 20,000 bits/s and each input sample is quantized into 1024 different voltage levels.

   i) What is the sampling frequency and the folding frequency?

   ii) What is the Nyquist rate for the signal $X_a(t)$?

   iii) What are different frequencies in the resulting discrete-time signals x[n]?

   iv) What is the resolution Δ? 7

   b) Highlighting the key features of Digital Signal processing, briefly differentiate Digital Signal Processor with Analog Signal Processor.

2. a) Define LTI system. Mention and explain the properties of LTI system. 8

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OR 7

Show that the multiplication of two DFT sequences results in circular convolution

b) Obtain a linear convolution of the following two discrete-time signals:

$x(n) = \{0,1,2,3\}$ and $h(n) = 2n[u(n)-u(n-3)]$

3. a) For the given system $y[n] = 0.7\, y[n-1] - 0.1\, y[n-2] + x[n]$, plot the pole zero diagram and magnitude response of the system 8

   b) Determine the causal signal x[n] having Z-Transform $X(z) = 1/\{(1-2z^{-1})(1-z^{-1})^2\}$. 7

4. a) Define DFT. Also compute 4-point DFT of the following sequence: 7

$$x(n) = \{1, 2, 3, 4\}$$

   b) . Find 8- point DFT of the sequence. {0,0,0,0,1,1,1,1,} using radix 2 DITFFT. 8

5. Consider here is given a three stage lattice filter with coefficient $K_1 = ¼$, 8

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$K_2 = ½$, and $K_3 = 1/3$, determine the FIR filter coefficient for the direct-form structure. 7

b) Determine the lattice coefficients corresponding to the FIR filter with system function and also draw the lattice structure.

$$H(Z) = 1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}.$$

6. a) Design an IIR filter by using Bi-linear transformation to meet the following specification $w_p = 0.25\pi$, $w_s = 0.55\pi$, $\alpha_s = 15$dB, $\alpha_p = 0.5$dB. 8

OR

Design a linear FIR filter using Kaiser window to meet the following

$$0.99 \leq |H(e^{j\omega})| \leq 1.01, for 0 \leq |\omega| \leq 0.19\pi$$

$$\leq |H(e^{j\omega})| \leq 0.01, for 0.21\pi \leq |\omega| \leq \pi$$

b) Draw the flowchart of Remez algorithm and explain it

OR

The transfer function of Analog filter is $Ha(s) = \frac{2}{(s+2)(s+4)}$ with

T = 0.1 sec. Design the IIR digital filter by using impulse invariance method.

7. Write short notes on (Any Two): 5×2

a) Gibb's Phenomena

b) Cascade structure for FIR system

c) ROC of Z transform

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NEPAL COLLEGE OF INFORMATION TECHNOLOGY

Assessment Fall 2025

Level: Bachelor
Programme: BE CE
Course: Digital Signal Analysis and Processing **(NEW)**

Year: 2025
Full Marks: 100
Pass Marks: 45
Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1 a) A digital communication link carries binary code words representing samples of an input signal The link is operated at 10.000 bits/sec and each input sample is quantized into 512 different voltage levels. 8

i. What is the sampling frequency and folding frequency?
ii. What is the Nyquist rate for the signal?
iii. What are the frequencies in resulting discrete time signal x(n)?
iv. What is the resolution?

b) Determine the convolution sum of two sequences using graphical method: 7

$$x(n) = \{1,2,3,4\}, h(n) = \{\underset{\uparrow}{1},2,3,1\}$$

2 a) Define Z transform. Find the Z transform 8

$$x[n] = a^n \sin(w_0 n) u[n]$$

Or

Find the causal signal x(n) of

$$X(z) = \frac{z^{-1}}{3 - 4z^{-1} + z^{-2}}$$

b) An LTI system is described by the equation 7

$$y(n) = x(n) + 0.51x(n-1) - 0.76x(n-2) - 0.45y(n-1) - 0.2y(n-2)$$

Determine the transfer function of the system. Sketch the poles and zeros on the z-plane. Assess the stability.

3 a) Obtain the cascade realization of the system 8

$$H(z) = \frac{3 + 2z^{-1} + z^{-2}}{[1 + (1/3)z^{-1}][1 - (1/3)z^{-1}][1 + (1/3)z^{-1}]}$$

b) Draw the lattice structure for the following IIR system. 7

$$H(z) = \frac{1}{1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}}$$

4 a) Convert the analog filter with system function 7

$$H(s) = \frac{s + 0.5}{(s + 0.5)^2 + 9}$$

Into a digital IIR filter by means of the bilinear transformation. Digital filter is to have a resonant frequency of $\omega_r = \pi/2$.

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b) Design a digital Butterworth low pass filter using Bilinear Transformation method to meet the following specifications: 8

Passband attenuation ≤ 1.28 dB
Passband edge frequency=200 Hz
Stopband attenuation ≥ 13db
Stopband edge frequency $= 350$ Hz
Sampling frequency= 1KHz.

Or

Design a Butterworth digital filter using the bilinear transformation. The specifications of the desired low-pass filter are:

$$0.8 \leq |H(\omega)| \leq 1;\ 0 \leq \omega \leq \frac{\pi}{2}$$

$$|H(\omega)| \leq 0.15;\ \frac{3\pi}{4} \leq \omega \leq \pi$$

with $T = 1$ s

5 a) How FIR digital filters are designed using different approaches? How would you use a rectangular window to design a FIR filter? Explain. 7

b) Design a linear FIR filter using Kaiser window to meet the following specifications: 8

$$0.99 \leq |H(e^{\wedge}jw)| \leq 1.01;\ \text{for } 0 \leq |w| \leq 0.19\pi$$

$$|H(e^{\wedge}jw)| \leq 0.01;\ \text{for } 0.21\pi \leq |w| \leq \pi$$

6 a) Find the 8-point DIFFFT of x(n) = {2, 2, 2, 2, 1, 1, 1, 1}. 8

b) Find the circular convolution of $x_1(n) = \{1,2,1,2\}$ and $x_2(n) = \{4,3,2,1\}$ by the graphical method. 7

7 Write short notes on: (**Any two**) 5 x 2=10

a) Causal System

b) properties of Z transform

c) Frequency response of LTI system

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Nepal Engineering College

Assessment (Fall Semester)

Year: 2025

Level: Bachelor
Program: Computer
Course: Digital Signal Analysis &Processing
Year/Part: III/I

Full Marks: 100

- ✓ *Candidates are required to give their answers in their own words as far as practicable.*
- ✓ *Assume suitable data if necessary or missing.*
- ✓ *Attempt all the questions*

1 a Define elementary signals. Explain and draw different types of basic elementary signals. 7

b A digital communication link carries binary-coded words representing samples of an input signal 8

$x_a(t) = 4\cos 3500\pi t + \sin 4500\pi t$.

The link is operated at 42,000 bits/s and each input sample is quantized into 1024 different voltage levels.

a) What is the sampling frequency and folding frequency?
b) What is the Nyquist rate for this signal?
c) What is the discrete-time signal obtained after sampling?
d) What is the resolution Δ?

OR

Define discrete-time system. Determine whether or not the given signal $y[n] = nx[2-n^2]$ is

i) Linear ii) Causal iii) stable iv) time-invariant v) with memory or not?

2 a State and prove time shifting and convolution properties of discrete-time Fourier transform. 7

b Consider an LTI system with impulse response $h[n] = \left\{1, \underset{\uparrow}{4}, 2, -1\right\}$ and input $x[n] = \{1, 1, \underset{\uparrow}{4}, 13, 15\}$. Determine the output of the system, y[n] using graphical method. 8

3 a Define z-transform and ROC. Determine the z-transform and sketch the Roc of the following signal (use z-transform properties). 7

$$x[n] = n^2 0.8^n u[n]$$

b Draw the pole-zero plot and find the magnitude response of the following system: 8

$$H(z) = \frac{1 + 2z^{-1}}{1 - 0.5z^{-1}}$$

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4 a Consider the following system described by LCCD equation: 7
y[n] = 0.6y[n-1] - 0.11y[n-2] + 0.3y[n-3] + x[n] + 2x[n-1] + 3x[n-3].
Draw the direct form- I and form-II structures.

b Determine the lattice coefficients corresponding to the FIR system with the system function 8

$$H(z) = 1 + \frac{7}{9}z^{-1} + \frac{3}{5}z^{-2} + \frac{1}{7}z^{-3}$$

5 a Design a lowpass Butterworth filter to meet the following specifications 8
Passband cutoff frequency: 200 Hz
Stopband cutoff frequency: 500 Hz
Passband ripple: 3 dB.
Stopband ripple: 21 dB.

OR

The z-transform of the signal is described as

$$X(z) = \frac{1 + 3z^{-1}}{(1 - 0.5z^{-1})(1 + 0.9z^{-1})}$$

What are the possible ROCs? Determine the inverse z-transform for causal signal.

b Differentiate FIR and IIR digital filters. 7

6 a Suppose we have two sequences x[n] and h[n] as follows: 7
x[n] = {1, 1, 4, -1, 7}; h[n] = {1, 4, 5, -2}.
Calculate circular convolution graphically.

b Find 8-point DFT of the sequence {12, 2, 5, 6, 0, -6, -5, -2} using radix-2 decimation in time domain. 8

7 Write short notes on (Any two): 5*2
a Causality and stability of LTI system
b Butterworth Filter
c Properties of z-transform

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Order of Filter N	Transfer function H(s) = 1/A(s) where A(s)
2	$(s^2+0.707s+1)$
3	$(s^2+s+1)(s+1)$
4	$(s^2+0.766s+1)(s^2+1.848s+1)$
5	$(s^2+0.618s+1)(s^2+1.618s+1)(s+1)$
6	$(s^2+0.518s+1)(s^2+4.414s+1)(s^2+1.932s+1)$
7	$(s^2+1.802s+1)(s^2+1.247s+1)(s^2+0.445s+1)(s+1)$

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United Technical College

Level: Bachelor | Semester-Fall | Year: 2025

Programme: BE | Full Marks: 50

Course: Digital Signal Analysis and Processing | Pass Marks: 25

Time: 1.5 Hours

Instructions:

Candidates are required to answer the questions in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt any three from 1-6 questions. Q.N.7 is compulsory.

Draw neat and labelled diagrams wherever necessary.

1. a. Define DSP. Explain the applications of DSP in the field of communication engineering. [7]

   b. A Digital communication link carries binary coded words representing samples of an input signal $x_a(t) = 3cos5000\pi t + 2sin6000\pi t$ Link is operated at 18,000 bits/s. and each input sample is quantized into 512 different voltage levels.

      i. What is the discrete time signal obtained after sampling?

      ii. What us the resolution? [8]

2. a. A discrete time Signal x(n) is applied to a discrete time LTI system with unit impulse h(n) . Find output or response y(n) if given that

   $x(n) = a^n u(n)$, $0 < a < 1$ and $h(n) = u(n)$ [7]

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   b. Define Z transform. Determine the Z transform of

   $x(n) = na^n u(n)$ [8]

3. a. Find the inverse z - transform of the system [7]

$$X(z) = \frac{5 - 2z^{-1}}{2 - z^{-1} - 1.5z^{-1}}$$

   For the following ROC

   i. $|z| < 1$ ii. $|z| > 3$ iii. $1 < |z| < 3$

   b. Obtain the circular convolution of the following sequences [8]

$$x_1(n) = \{1, 2, 1, ,1\} and\ x_2(n) = \{4, 0, 2, 2\}$$

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4. a. What is the importance of FFT? Find 8-point DFT of sequence $x(n) = \{1, 2, 0, 5, 3\}$ using DIT FFT algorithm [7]

b. Obtain cascade and parallel structure for the following system
$y(n) = y(n-1) - 0.5y(n-2) + x(n) - x(n-1) + x(n-2)$ [8]

5. a. Draw lattice and ladder structure [7]

$$H(z) = \frac{1 - z^{-1} + 0.5z^{-2}}{1 + 0.2z^{-1} - 0.15z^{-2}}$$

b. Design an FIR filter to meet the following specifications [8]
Pass band = 2 kHz
Stop band = 5 kHz
Stop band attenuation = 42 dB
Sampling frequency = F_s = 20 kHz
Use Hamming Window

6. a. Determine H(z) using impulse invariance method for the following system function

$$H_a(s) = \frac{s+0.1}{([s+0.1])^2 + 3^2}$$ [7]

b. Using bilinear transformation design a Butterworth filter which satisfies the following conditions

$0.8 \leq |H(e^{j\omega})| \leq 1$ $\quad 0 \leq w \leq 0.2\pi$

$|H(e^{j\omega})| \leq 0.2$ $\quad 0.6\pi \leq \omega \leq \pi$ [8]

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7. Write shorts note on **(any one)** [5]
   a. Computational Complexity of FFT
   b. Recursive and non-recursive system
   c. Remez Exchange Algorithm

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Universal Engineering College

Level: Bachelor | Semester-Fall | Year: 2025
Programme: BE | Full Marks: 100
Course: Digital Signal Analysis and Processing **(NEW)** | Pass Marks: 45
Time: 3hrs

Candidates are required to give their answers in their own words as far as practicable
The figures in the margin indicate full marks
Attempt all the questions.

1 a) A digital communication link carries binary code words representing samples of an input signal The link is operated at 10 000 bits/sec and each input sample is quantized into 512 different voltage levels 8
i What is the sampling frequency and folding frequency?
ii What is the Nyquist rate for the signal?
iii What are the frequencies in resulting discrete time signal x(n)?
iv What is the resolution?

b) Determine the convolution sum of two sequences using graphical method 7

$$x(n) = \{1,2,3,4\}, h(n) = \left\{\underset{\uparrow}{1},2,3,1\right\}$$

2 a) Define Z transform. Find the Z transform 8

$$x[n] = a^n \sin(w_0 n) u[n]$$

Or

Find the causal signal x(n) of

$$X(z) = \frac{z^{-1}}{3 - 4z^{-1} + z^{-2}}$$

b) An LTI system is described by the equation 7
$y(n) = x(n) + 0.51x(n-1) - 0.76x(n-2) - 0.45y(n-1) - 0.2y(n-2)$
Determine the transfer function of the system. Sketch the poles and zeros on the z-plane. Assess the stability.

3 a) Obtain the cascade realization of the system 8

$$H(z) = \frac{3 + 2z^{-1} + z^{-2}}{[1 + (1/3)z^{-1}][1 - (1/3)z^{-1}][1 + (1/3)z^{-1}]}$$

b) Draw the lattice structure for the following IIR system. 7

$$H(z) = \frac{1}{1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}}$$

4 a) Convert the analog filter with system function 7

$$H(s) = \frac{s + 0.5}{(s + 0.5)^2 + 9}$$

Into a digital IIR filter by means of the bilinear transformation. Digital filter is to have a resonant frequency of $\omega_r = \pi/2$

b) Design a digital Butterworth low pass filter using Bilinear Transformation method to meet the following specifications 8
Passband attenuation ≤ 1.28 dB
Passband edge frequency=200 Hz
Stopband attenuation ≥ 13db
Stopband edge frequency = 350 Hz
Sampling frequency= 1KHz

Or

Design a Butterworth digital filter using the bilinear transformation. The specifications of the desired low-pass filter are

$$0.8 \le |H(\omega)| \le 1;\ 0 \le \omega \le \frac{\pi}{2}$$

$$|H(\omega)| \le 0.15;\ \frac{3\pi}{4} \le \omega \le \pi$$

with $T = 1$ s

5 a) How FIR digital filters are designed using different approaches? How would you use a rectangular window to design a FIR filter? Explain. 7

b) Design a linear FIR filter using Kaiser window to meet the following specifications 8

$$0.99 \le |H(e^{jw})| \le 1.01;\ \text{for } 0 \le |w| \le 0.19\pi$$

$$|H(e^{jw})| \le 0.01;\ \text{for } 0.21\pi \le |w| \le \pi$$

6 a) Find the 8-point DIFFFT of x(n) = {2, 2, 2, 2, 1, 1, 1, 1} 8

b) Find the circular convolution of $x_1(n) = \{1,2,1,2\}$ and $x_2(n) = \{4,3,2,1\}$ by the graphical method 7

7 Write short notes on: (Any two) 5 x 2 = 10
a) Causal System
b) properties of Z transform
c) Frequency response of LTI system

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COSMOS COLLEGE
OF MANAGEMENT AND TECHNOLOGY

Term Test II

Date: 2082/11/10		Full Marks	50
Level	BE	Time	
Programme	BCE		1.5 hrs
Semester	V		

Subject: *Digital Signal Analysis and Processing*

- ✓ *Candidates are required to give their answers in their own words as far as practicable.*
- ✓ ***Attempt All question.***
- ✓ *The figures in the margin indicate* ***Full Marks***.
- ✓ *Assume suitable data if necessary.*

1 a) Why Direct form II is better than Direct form I? Draw the direct form realization of the following difference equation. 8

$$2y(n) + y(n-1) + 4y(n-3) = x(n) + x(n-1)$$

b) Obtain the lattice ladder structure of the discrete time system described by the difference equation. 7

$$y(n) = -\frac{3}{4}y[n-1] + \frac{1}{4}y[n-2] = x[n] + \frac{1}{2}x[n-2]$$

2 a) Design a digital low pass Butterworth filter by applying bilinear transformation technique for the given specification. 8

Passband edge = 120 Hz
Passband attenuation = 1 dB
Stopband edge = 170 Hz
Stopband attenuation = 16 dB . Assume sampling frequency of 512 Hz.

b) Obtain H(z) using the impulse invariance techniques for an analog system function which is given by: 7

$$H(S) = \frac{1}{(S+0.5)(S^2+0.5\,S+2)}$$

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3 a) Why Kaiser window is better than other windows? How can you design FIR filter using Kaiser window? Explain with necessary mathematical relations. 8

b) How can you design FIR filter using Frequency sampling method? Explain. 7

4 Write Short notes on: (Any One) 5

a) Remez Exchange Algorithm

b) Frequency Warping

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LUMBINI ENGINEERING MANAGEMENT AND SCIEMCE COLLEGE

Level: Bachelor | Year: 2026
Programme: BE | Full Marks: 100
Course: Digital Signal Analysis and Processing | Pass Marks: 45
Semester:-5th | Time: 3 hrs.

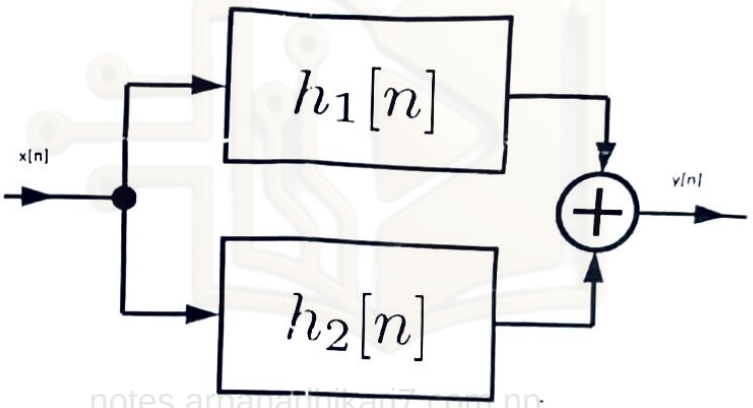
Candidates are required to give their answers in their own words as far as practicable.
The figure in the margin indicates full marks.
Attempt all the questions.

1.

a. A digital communication link carries binary coded words representing samples of an input signal $x_a(t) = 5\cos 600\pi t + 7\cos 800\pi t$. The link is operated at 1000 bits/sec and each input sample is quantized into 1024 different voltage levels. [2+1+2+2]

i. What is the sampling frequency and folding frequency?
ii. What is the Nyquist rate for the signal $x_a(t)$?
iii. What are the frequencies in the resulting discrete time signal x[n]?
iv. What is the resolution 'Δ'?

b. Two subsystems $h_1[n]$ and $h_2[n]$ are interconnected as shown in the block diagram. Determine the response of the system if; $h_1[n] = \{1,4,2\}$ and $h_2[n] = \{2, 3, 1\}$, when excited by input $x[n] = \{2, 4\}$. [8]



2.

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a. Prove that a discrete time LTI system is stable if and only if it's impulseresponse is absolutely summable. Determine whether the given discrete time system described by LCCDequation is [5+2]

i. Time invariance

$$y[n] = -2y^2[n-1] + 3x[n] + 2x[n-1]$$

b. Determine the inverse Z-transform of [8]

$$X(Z) = \frac{1}{1 - 1.5z^{-1} + 0.5z^{-2}}$$

i. ROC: $|z| > 1$
ii. ROC: $|z| < 0.5$
iii. ROC: $0.5 < |z| < 1$

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Also specify the causality and stability in each case.

3

a. Compute the 4-point DFT of the sequence
$Xa(t) = 4cos\ 200\pi t$, with sampling frequency of 500 Hz. 7

b. Determine the response of the system using FFT algorithm, if the input x[n]and impulse response h[n] are given as under; 8
X[n]={2, 2, 4} and h[n]={1, 1}

4.

a. Determine the cascade and parallel realization of the discrete time systemdescribed by differential equation. [8]

$$y[n] = -\frac{3}{4}y[n-1] + \frac{1}{4}y[n-2] + x[n] + \frac{1}{2}x[n-1]$$

b. Obtain the lattice ladder structure of the discrete time system described bythe differential equation [7]

$$y[n] = -\frac{3}{4}y[n-1] + \frac{1}{4}y[n-2] + x[n] + \frac{1}{2}x[n-1]$$

Also check the stability of the filter

5.

a. Design a digital low pass Butterworth filter by applying bilinear transformation technique for the given specifications. [8]
Pass band edge = 120Hz
Pass band attenuation = 1dB
Stop band edge = 170Hz
Stop band attenuation = 16 dB Assume
sampling frequency of 512 Hz

b. Obtain H(z) using the impulse invariant techniques for an analog system function which is given by: [7]

$$H_a(s) = \frac{1}{(s+0.5)(s^2+0.5\,s+2)}$$

6.

a. Design a low pass digital filter to be used in A/D and D/A structure that will have -3 dB cut-off at $30\pi\ rad/sec$ and an attenuation of 50 dB at
$45\,\pi\frac{rad}{sec}$, the filter is required to have a linear phasse and the system uses sampling rate of 100 samples/second. [7]

b. Design an FIR linear phase filter using Kaiser window to meet the following specifications:

$$0.99 \leq |H(e^{i\omega})| \leq 1.01, \quad \text{for } 0 \leq |\omega| \leq 0.19\pi,$$
$$|H(e^{i\omega})| \leq 0.01, \quad \text{for } 0.21\pi \leq |\omega| \leq \pi$$

7. Write short note on(Any two) 2*5=10
a. Recursive and non-recursive system
b. FIR Vs IIR
c. Circular Convolution Vs linear convolution

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Pokhara University

Everest Engineering College

Final Internal Assessment
Fall-2025

Level: Bachelor — **F.M.** 100
Program: BE – CMP(5th Sem) — **P.M.** 45
Section: — **Time:** 3hrs
Subject: Digital Signal Analysis and Processing

Attempt all the questions.

1 a. A digital Communication link carrier binary –coded word representing samples of an input signal. 8

$$x_a(t) = 3Cos700\pi t + 2Sin1400\pi t$$

The link operated at 18000 bits/s and each input sample is quantized into 1024 different voltage levels.

a. Determine the Sampling frequency?

b. What is the Nyquist rate of the signals?

c. What is the discrete-time Signal obtained after sampling?

d. what is resolution Δ? 7

b. Define the Energy and Power signal. Compute the Energy and power of unit Impulse Signal.

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2 a) Find the Convolution between two signal 7

$$x[n] = u[n] - u[n-3] + \delta[n-3] \text{ and } h[n] = \left(\frac{1}{3}\right)^n (u[n] - u[n-2])$$

verify the answer and Draw final result in graphical method

b) Define Fast Fourier Transform (FFT)? Determine 8 –point DFT of the sequence Using Decimation in Time FFT. $x[n] = \cos(\frac{n\pi}{4})$ 8

3 a. Show that the Multiplication of Two DFT Sequence results in Circular Convolution. 8

b. Find the Z – Transform of the given Signal $x[n] = a^n Sinw_o nu[n]$. 7

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4 a. Design a digital low pass Butterworth filter by applying Bilinear transformation technique for the given specifications. 8

Pass band edge = 120 Hz pass band attenuation = 1dB

Stop band edge = 170 Hz stop band attenuation = 16 dB

Assume Sampling frequency of 512 Hz

b) Draw lattice ladder of the discrete time System described by the Differential equation. 7

$$y[n] = -\frac{3}{4}y[n-1] + \frac{1}{4}y[n-2] + x[n] + \frac{1}{2}x[n-1]$$

5 a. Obtain H(z) Using the Impulse invariant techniques for an analog System Function which is given by 7

$$H_a(s) = \frac{1}{(s+0.5)(s^2+0.5s+2)}$$

b. Determine the Cascade and parallel Realization of H(z) of question 5(a) 8

6 a. Design a low pass FIR System to meet the following Specifications. 8

Passband edge Frequency = 2KHz Stopband edge frequency = 5KHz

Stopband attenuation = 42dB Sampling frequency = 20KHz

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b. For a System with Poles at 0.45 ± j1.06 and Zeros at 0.58 ± j2.06. Plot the magnitude and frequency response of the System. 7

7 Write short notes on: (Any two) **2*5=10**

a) Linear Convolution Vs Circular Convolution

b) Remez Exchange Algorithm

c) Properties of Z-Transform

Good Luck

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GANDAKI COLLEGE OF ENGINEERING AND SCIENCE

Level: Bachelor Semester: Fall Year : 2025
Programme: BE CE V Full Marks: 100
Course: **Digital Signal Analysis and Processing** Pass Marks: 45
Time : 3hrs.

The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Differentiate between energy and power signal. Determine whether these signals are linear or not. 7

a) $y(n) = ax(n)+6$ b) $y(n) = x(n)+nx(n+1)$

OR

Define signal. Explain basic elements of DSP with block diagram.

b) Verify using commutative property of convolution for the $x(n) = \{\mathbf{1},2,12\}$, $h(n) = \{2,2,\mathbf{-1},1\}$. Here bold number represents value at origin. Also sketch the given signals and output response graphically 8

2. a) A causal LTI system is described by the difference equation: $y(n) - y(n-1) + y(n-2) + x(n) + 2x(n-1)$. Find the system function and frequency response of the system. Plot the poles and zeros and indicate the ROC. 8

b) Determine the inverse Z-transform, when, ROC: $|z| > 1$ 7

ROC: $|z| < \frac{1}{2}$

$$X(z) = \frac{1}{1 + \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}}$$

3. a) Obtain direct form I and direct form II for the following system. $y(n) = 0.1y(n-1) + 0.72y(n-2) + 0.7x(n) - 0.252x(n-2)$ 7

b) Draw the lattice structure for the following FIR filter function 8

$$H(z) = 1 + 2z^{-1} + \frac{1}{3}z^{-2}$$

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4. a) The transfer function of an analog filter is, $H_a(s)=2/((s+2)(s+4))$ with sampling period T = 1 sec. Design the IIR digital filter using the impulse invariance method.

b) Design a digital lowpass Butterworth filter to meet the following specifications using bilinear transformation method. $|H(ej\omega)| = 0.8; \quad |\omega| \leq 0.25\pi$ $|H(ej\omega)| = 0.25; \quad 0.45\pi < |\omega| < \pi$ 8

5. a) Explain the FIR filter design steps using Kaiser window. 7

b) Find the 8-point DIT-FFT of $x(n)=\sin(\pi n/3)$. 8
Hint: take n =0,1,.....7

OR

Compute the 8-point DFT of the sequence x(n) = (2,2,2,2,1,1,1,1) Using radix-2 DIF FFT algorithm:

6. a) Design a filter with $H_d(e^{jw}) = \begin{cases} 1 & -\frac{\pi}{6} \leq w \leq \frac{\pi}{6} \\ 0 & otherwise \end{cases}$ 7
Using a Hanning window with M = 9.

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b) Draw the lattice ladder structure of following system. 8

$$H(z) = \frac{1-0.8z^{-1}-0.9z^{-2}}{1+0.2z^{-1}+0.8z^{-2}}.$$

7 Write short notes on **any two**: 5x2

a) Linear Convolution vs Circular Convolution

b) Linear constant coefficient difference equation

c) DTFT and properties

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POKHARA UNIVERSITY
FACULTY OF SCINCE AND TECHNOLOGY
SCHOOL OF ENGINEERING

Exam	Final Internal Examination 2025		
Level	B.E.	F M	100
Program	BE Computer	PM	45
Year/ Part	III/I	Time	3 Hrs

Subject: Digital Signal Analysis and Processing

Candidates are required to give answers in their own words as far as practicable.
The figure in the margin indicates full marks. Assume suitable data if necessary.
Attempt all the questions.

1.	a) What is digital Signal processing? Name the basic elements of digital signal processing and briefly explain each of them. List any two advantages of Digital signal processing.	8
	b) Given a discrete time signal $x[n] = \{1, 1, 1, 1, \frac{1}{2}, \frac{1}{2}\}$, i. sketch $x[n]$, ii. Sketch $x[2-n]$ and iii. Determine $x[n]u[2-n]$	7
2.	a) Verify the commutative property of convolution where input signal, x[n] = {1, 1, 1, 1} and response of system, h[n] = {1, 1, 1, 1}.	8
	b) What are the two major characteristics of ROC? Determine the inverse z-transform of: $H(z) = \frac{1+2z^{-1}+z^{-2}}{1+4z^{-1}+4z^{-2}}$ (choose any one of the methods)	7
3.	a) Define DFT and IDFT. Explain DFT as a linear transformation.	8
	b) Find the circular convolution of the sequence: $x(n) = \{0,1,2,3\}$ $h(n) = \{2\ 1,1,2\}$	7
4.	a) Determine the 4-point DFT of the following sequence using DIF FFT radix-2 algorithm: $x(n) = u(n) - u(n-4)$	8
	b) Draw the lattice structure from the given FIR filter's system function: $H(z) = 1 + \frac{13}{12}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}$ **OR** Given a 3-stage lattice filter with coefficient $K_1 = \frac{1}{4}, K_2 = \frac{1}{2}$ *and* $K_3 = \frac{1}{3}$. Draw the corresponding FIR filter direct form I or II model.	7
5.	a) Design a normalized linear phase FIR filter having the phase delay of 4 and attenuation of at least 40 dB in the stopband. Also obtain the magnitude response of the filter. **OR** Design a linear FIR phase filter using kaiser window to meet the following specifications: - $0.99 \le \lvert H(e^{iw}) \rvert \le 1.01, for\ 0 \le \lvert w \rvert \le 0.19\pi$	8

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	$\lvert H(e^{jw})\rvert \leq 0.01, for\ 0.21\pi \leq \lvert w\rvert \leq \pi$ **b)** Design a lowpass FIR filter using frequency sampling technique having cut off frequency of $\frac{\pi}{2}$ rad/sample. The filter should have linear phase of 4 and length of 17.	7
6.	**a)** The transfer function of an analog lowpass filter is $H_a(s) = \frac{1}{s+1}$, and its bandwidth is 1 rad/sec. Design the digital IIR filter using Bilinear transformation method whose cut-off frequency is 0.2π and sampling interval is 0.0167 sec. **b)** Find the order and cut-off frequency of a digital filter by using impulse invariance method with the following specifications: $0.89 \leq \lvert H(e^{jw})\rvert \leq 1$, for $0 \leq w \leq 0.4\pi$ $\lvert H(e^{jw})\rvert \leq 0.18$, for $0.4\pi \leq w \leq \pi$ Also, plot its poles.	7 8
7.	**Write short notes on:** *(Any two)* **a)** Necessary and sufficient condition for a system to be stable **b)** Gibbs Phenomena in FIR filter design. **c)** LTI system	2×5

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*** ***Best of Luck*** ***

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Madan Bhandari College of Engineering
Urlabari-3, Morang
Final Internal Examination

Level: Bachelor of Computer Eng. **Full Marks: 100**
Programme: BE **Pass Marks: 45**
Year/Part: III/I **Time: 3 hrs**
Subject: - DSAP

1. Explain the basic element of DSP system with a net block diagram. Discuss the at least there practical application of DSP system. (7)
2. Determine whether the unit step signal is energy or power signal or neither. (5)
3. Find the output of LTI system having impulse response $h(n) = (\frac{1}{2})^n [u(n+2)-u(n-2)]$ to the input $x(n) = [2,1,0,-1,4]$. (5)

4. a) State and prove the parseval's theorem of the z- transform and explain its physical significance. (7)
   b) Determine the Z-transform and ROC of the signal $x(n) = \alpha^n u(n)$ for $0<\alpha<1$.

5. Given the analog signal $X(t)= 3 \cos 2000\pi t +5\sin 6000\pi t+10\cos 12000\pi t$ and find the following. (10)
   a) What is the Nyquist rate?
   b) What is the discrete time signal after samping , sampling rate is 5000 sample /sec.
   c) What is the data rate if each sample signal is quantised 1024 different voltage level?
   d) Find the resolution.
   e) What is the analog signal that can be reconstructed from samples.
6. Find the circular convolution of the sequence $x_2(n) = [1,2]$ and $x_2(n)= u(n)-u(n-4)$ in graphical method. (5)

7. a) Draw the direct-I and direct-II structure of following. (10)
   $y[n]-0.75y[n-1]-0.25y[n-2] = x[n]+x(n-1)$
   b) Compute lattice coefficient.

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8. Given $x(n)=\{0,1,2,3\}$ find x(k) using DIT-FFT algorithm (5)
9. Determine the system function of a digital filter H(z) using impulse invariance method at sampling frequency from $H_a(s)$ as given below. (8)
   $$H_a(S) = \frac{2}{(s+1)(s+2)} \quad \text{and } fs = 2Hz.$$
10. Write the algorithm of chebyshev approximation of low pass filter. (10)
11. Design a second order discrete time Butterworth filter with cut off frequency of 1 KHz and sampling frequency of 10^4 samples/sec by using bilinear transformation method. (10)

12. Design the linear phase FIR fiter using Kaiser window meet the following specification. (10)
    $0.99 \le |(He^{jw})| \le 1.01$ for $0 \le w \le 0.91\pi$
    $|(He^{jw})| \le 0.01$ for $0.21\pi \le w \le \pi$

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Madan Bhandari College of Engineering
Urlabari-3, Morang
Final Internal Examination

Level: Bachelor of Computer Eng. | Full Marks: 100
Programme: BE | Pass Marks: 45
Year/Part: III/I | Time: 3 hrs

Subject: - DSAP (Set-B)

1. Explain the basic element of DSP system with a net block diagram. Discuss the at least there practical application of DSP system. (7)
2. Determine whether the unit ramp signal is energy or power signal or neither. (5)
3. Find the output of LTI system having impulse response $h(n) = (½)^n [u(n+2)-u(n-2)]$ to the input $x(n) = [1,2,0,4,0]$. (5)

4. a) State and prove the parseval's theorem of the z- transform and explain its physical significance. (7)
   b) Determine the Z-transform and ROC of the signal $x(n) = \alpha^n u(n)$ for $0<\alpha<1$.

5. Given the analog signal $X(t)= 5\cos 4000\pi t + 4\sin 8000\pi t + 2\cos 6000\pi t$ and find the following. (10)
   a) What is the Nyquist rate?
   b) What is the discrete time signal after samping , sampling rate is 2000 sample /sec.
   c) What is the data rate if each sample signal is quantised 1024 different voltage level?
   d) Find the resolution.
   e) What is the analog signal that can be reconstructed from samples?
6. Find the circular convolution of the sequence $x_1(n) = [2,1]$ and $x_2(n) = u(n)-u(n-4)$ in graphical method. (5)

7. a) Draw the direct-I and direct-II structure of following. (10)
   $y[n]-0.75y[n-1]-0.25y[n-2] = x[n]+x(n-1)$
   b) Compute lattice coefficient.

8. Given $x(n)=\{0,1,0,2\}$.find $x(k)$ using DIT-FFT algorithm. (5)
9. Determine the system function of a digital filter H(z) using impulse invariance method at sampling frequency from $H_a(s)$ as given below. (8)

$$H_a(S) = \frac{2}{(s+1)(s+2)} \quad \text{and } fs = 4Hz.$$

10. Write the algorithm of chebyshev approximation of low pass filter. (10)
11. Design a second order discrete time Butterworth filter with cut off frequency of 1-KHz and sampling frequency of 10^4 samples/sec by using bilinear transformation method. (10)

12. Design the linear phase FIR fiter using Kaiser window meet the following specification. (10)

$0.99 \le |(He^{jw})| \le 1.01$ for $0 \le w \le 0.91\pi$
$|(He^{jw})| \le 0.01$ for $0.21\pi \le w \le \pi$

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